Inverse Airfoil Design as Constrained Root-Finding: A Monolithic CST–Newton Formulation

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Abstract

We recast inverse airfoil design – recovering a geometry that produces a prescribed surface pressure or edge-velocity distribution – as a single determined nonlinear root-find rather than an objective-function search. Parameterising the surface with Class-Shape-Transformation (CST) coefficients that enter the geometry *linearly* makes the geometric Jacobian block exact, constant, and design-independent, and makes geometric design constraints (leading-edge radius, trailing-edge thickness, inscribed area) linear algebraic rows rather than nonlinear predicates. Appending the CST coefficients as unknowns to a coupled viscous/inviscid Newton solver (mfoil) therefore converts constrained shape *optimization* into constrained *root-finding*: one square system, no outer loop, no surrogate. We validate the architecture with a falsifiable self-consistency test – recovering a known CST coefficient vector from its own self-generated target – to $\|A - A^*\|_\infty = 1.079 \times 10^{-11}$ in 4 Newton iterations with a visible quadratic tail, and confirm the recovered geometry reproduces the target in fully released (natural-transition) direct mode to $\Delta c_l = 8.0 \times 10^{-13}$. An ablation matrix identifies the primary uniqueness guard for the resulting square system: sensitivity-optimal (QR-pivoted) selection of target stations, not initial-guess quality, is what separates recovering the true design from converging cleanly to a spurious but residual-zeroing root. Measured against a competently-tuned nested Levenberg–Marquardt baseline under two independent fair-paired controls, the monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations – a real but modest reduction, not the two-to-three-orders-of-magnitude headline hypothesised a priori – while retaining determinism, guaranteed local quadratic convergence, exact analytic constraint-row Jacobians, and per-iteration failure-mode diagnostics that the nested baseline has no analogue for. A subsequent 20-section NACA panel evaluation, run at the winning configuration the ablation matrix identifies, recovers all 18 generable sections to err_free_inf $\leq 1.5 \times 10^{-10}$; we report this alongside the interval-arithmetic reason its formal pre-registered Wilson-confidence criterion still fails at this panel size. We further place the architecture on a comparison table against MISES's own Modal-Inverse mode, the nearest prior CST-based inverse method, and the current generative/learned-surrogate inverse-design literature, on formulation class, cost, determinism, and constraint-handling grounds rather than a single flow-solve number, since the methods are not commensurable on that axis alone.

1 Introduction

Inverse aerodynamic design – given a target surface pressure or isentropic Mach distribution, recover the geometry that produces it – is classically posed either as a well-posed but restrictive conformal-mapping problem [Lighthill, 1945, Volpe and Melnik, 1986], or as an outer optimization loop wrapped around a converged forward flow solve. The latter is the practical default in industry: a geometry parameterisation, a surrogate or gradient-based optimizer, and hundreds

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to thousands of fully converged flow solves per design. The present paper's motivation is autobiographical: the author's own doctoral work on sCO_2 axial-turbine blading used exactly this pattern – Class-Shape-Transformation (CST) geometry, a B-spline control-point wrapper, and a Dakota/Kriging surrogate loop against MISES – and its future work section named the open problem verbatim: integrating CST directly with a Newton-based flow solver to escape the nested loop (docs/CST_MISES_Monolithic_Inverse_Design.md §1.4).

That escape route already exists in one solver. MISES's Modal-Inverse mode [Drela, 1990] appends mode-shape amplitudes directly to its global Newton system and drives them so that computed surface speed matches a target – a determined root-find, not a search. This is not new; Drela described it in 1986–1990. What MISES's built-in mode shapes are *not*, however, is a complete, geometrically meaningful, analytic design basis: they are heuristic bump functions, not a family whose coefficients double as leading-edge radius, trailing-edge thickness, or inscribed area. Kulfan's CST parameterisation [Kulfan, 2008] is such a basis, and Masters et al. [Masters et al., 2017] showed it is among the most efficient analytic families across a 2000-airfoil comparison. CST has also already been coupled to inverse design once, by Morris, Allen & Rendall [Morris et al., 2010]: there, CST is used to *smooth* a pressure-residual-driven normal-direction shape correction inside an outer iterative loop – CST is the smoother, not an unknown in a Newton system. The present work does not repeat either of these; it combines a property of CST that neither prior use exploits.

1.1 Related work

Classical inverse design. Inverse aerofoil design predates any of the parameterisations discussed below. Lighthill's conformal-mapping method [Lighthill, 1945] is the founding well-posed formulation: a target speed distribution on the unit circle is mapped conformally to the physical aerofoil, with three integral *closure conditions* – closure of the contour, a fixed leading-edge radius, and a prescribed circulation – required for a solvable, self-consistent shape to exist at all. This closure-condition structure is the direct conceptual ancestor of §2.2's linear rows: both are answers to the same question, "what makes an arbitrarily chosen target realisable by *some* member of the design family," asked eighty years apart under very different geometric representations. Volpe & Melnik [Volpe and Melnik, 1986] sharpened well-posedness for the transonic regime, showing that an inverse problem posed with pressure (rather than speed) as the target and with the wrong closure conditions is generically *ill*-posed – existence and uniqueness are not automatic once the target is chosen independently of the geometry family, a caveat that motivates the identifiability discussion of §5.1 in a completely different (discrete, finite-dimensional) setting. Eppler's design method [Eppler and Somers, 1980] and Selig & Maughmer's PROFOIL [Selig and Maughmer, 1992] are the mature, still-used engineering descendants: both are multi-point conformal-mapping inverse solvers, target several operating conditions' pressure distributions simultaneously by blending mode functions, and both remain outside the Newton-coupled, viscous-inclusive family this paper and MISES belong to – they close the inverse problem analytically at the potential-flow level, then couple to a boundary-layer method non-simultaneously, rather than solving flow and geometry as one system.

Parameterisation comparison. Sobieczky's PARSEC [Sobieczky, 1999] parameterises an aerofoil by twelve directly-meaningful quantities (LE radius, thickness/camber extrema and their locations, TE angle and thickness) fitted by a polynomial basis – geometrically transparent in the same spirit as CST's endpoint identities (Eq. (6)), but through a basis whose coefficients are not independent design freedoms in the same linear sense: PARSEC's basis functions depend on the target values themselves (the fit is re-solved per target), whereas CST's Bernstein basis $S_i(\psi)$ is fixed once n is chosen, independent of what shape is being represented – the property Eq. (5) depends on. Masters et al.'s benchmark [Masters et al., 2017] is the field's reference comparison: over 2000 aerofoils and several parameterisation families (CST, PARSEC, B-splines, Bezier, orthogonal/SVD/POD modes derived from a training database), CST is found to be among the most geometrically efficient *analytic* (i.e. not data-driven) families, needing roughly 10–14 coefficients per surface for sub-0.1%-chord fitting error on typical sections, while purely data-driven SVD/POD modes fit an existing database more efficiently still at the cost of needing that database and losing CST's closed-form geometric-functional structure (§2.2) entirely. This paper's choice of CST is therefore not merely convenience: it is the parameterisation Masters et al.'s own comparison identifies as the best-supported analytic basis, and the one whose particular algebraic property (linearity in A) this paper's contribution depends on.

Newton-coupled viscous/inviscid solvers. The architecture this paper appends CST to is Drela & Giles's ISES lineage [Drela and Giles, 1987]: a two-equation integral boundary-layer closure coupled *simultaneously* (not iteratively) to a panel-method inviscid solution through a single global Newton system, with the e^n transition model providing the closure that makes laminar-to-turbulent transition a smooth part of that system rather than a switched boundary condition. mfoil [Fidkowski, 2022] is a modern, open, from-scratch reimplementation of exactly this architecture – the ISES/XFOIL viscous-inviscid coupling without XFOIL's Fortran legacy code, in Python, with an exposed sparse global Jacobian – which is precisely why it is the Stage 1 testbed here (§2): the extended system of §2.3 could not be built without an analysis code that already exposes its own $\partial R/\partial U$ block for direct augmentation. Section 5.5 of the dossier (docs/CST_MISES_Monolithic_Inverse_Design.md) additionally documents that MISES's Parametric-Inverse mode was designed from the outset to accept an arbitrary geometry-definition system via a black-box interface (`BPREAD`/`BPWRIT`/`BLDGEN`), i.e. the mechanism this paper's Stage 3 (§7) would use already exists in the target solver and has simply never been exercised with a CST-as-unknowns formulation.

Modern learned inverse-design methods. A distinct, much newer line of work learns the pressure-to-geometry map directly from data rather than solving a physics-based system at all. Conditional GANs [Yilmaz and German, 2020] and, more recently, diffusion models – a classifier-free-guided denoising diffusion model reporting a 34.4% precision improvement over WGAN baselines on an airfoil-generation benchmark [Yu et al., 2025] – learn a stochastic generative map from a target Cp (or aerodynamic-coefficient) vector to a geometry, trained offline on a large precomputed database, with inference reduced to a single (or few-step) forward network pass. A hybrid approach, LF-PINN [Wang et al., 2024], trains a physics-informed surrogate directly on mfoil's own low-fidelity flowfields – the same solver used here – and reports at least an order-of-magnitude online inference speedup relative to running mfoil directly, again after an offline training investment. These methods answer a different question well: given a large library of prior designs and target statistics resembling that library, produce a plausible geometry near-instantly. They do not provide a convergence guarantee, an exact constraint-row Jacobian, or a certificate that the returned geometry actually reproduces the requested Cp to any specified tolerance – the network's output is evaluated *after* the fact by running a flow solver on it, not constrained *during* generation the way Eq. (12)'s equality rows are. §5.8's comparison table places this paper's architecture and the learned-inverse family on common columns (flow solves per design, determinism, constraint handling) explicitly, because the two are not competing solutions to the same technical problem so much as different points on an accuracy-guarantee vs. amortised-cost trade surface.

Turbomachinery inverse design. Lavagnoli [Lavagnoli, 2026] benchmarks Kolmogorov–Arnold Networks against MLP and Gaussian-process regressors for inverse turbine-passage design, trained on ≈30,000 blade profiles generated by an automated MISES-coupled optimisation pipeline (inlet flow angle $-50°$ to $0°$, outlet $50°$–$75°$); the trained network infers a loading-consistent geometry in ≈0.1 s with a mean outlet-angle error of $0.086°$. Structurally this is the same learned-surrogate pattern as the previous paragraph, specialised to cascades, and it is the closest published turbomachinery analogue of this paper's motivating problem (§1) – but it is once more amortised-cost statistical inference over a precomputed design database, not a per-target root-find with a convergence certificate. Notably, Lavagnoli's pipeline includes a predicted-RMS *feasibility proxy* that flags statistically ill-posed inverse targets before training-set generation – the statistical-learning analogue of this paper's realisability guard (§2.7), independently motivated by the same underlying problem (some targets are simply not achievable by the design family in use) but answered by a learned classifier rather than a closed-form linearisation.

The precise novelty claim. CST's surface parameterisation

$$\zeta(\psi) = C(\psi) \sum_i A_i S_i(\psi) + \psi\, \zeta_T \tag{1}$$

is *linear* in the coefficient vector A. Two consequences follow, and neither is available to a generic (e.g. B-spline-with-wrapper) geometry representation once local-frame perturbations, LE tangent-angle displacements, or re-fitting are introduced to reach the same design freedom (docs/CST_MISES_Monolithic_
(1) the geometric Jacobian block $\partial\zeta/\partial A_i = C(\psi) S_i(\psi)$ is exact, constant, and design-independent – assembled once, never re-derived inside a Newton loop; and (2) the geometric functionals that matter for manufacturability (LE radius, TE thickness, wedge angle, inscribed area, even leading-order curvature continuity) are *linear* in A, so they enter the Newton system as constant-coefficient algebraic rows rather than non-smooth predicates or penalty terms. The contribution of this paper is **not** "geometry unknowns in a Newton system" (that is Drela, 1986/1990) and **not** "CST used somewhere inside inverse design" (that is Morris, Allen & Rendall, 2010). It is the combination: an analytic basis whose linearity makes the geometric Jacobian exact and constant, *and* whose geometric functionals make design constraints linear rows, together converting *constrained shape optimization* into *constrained root-finding* – one square Newton system, solved once, no outer loop.

The honest headline. A priori, the architecture was hypothesised to require "two to three orders of magnitude fewer flow solves, plus determinism and uniqueness" relative to a nested-optimization baseline (docs/CST_MISES_Monolithic_Inverse_Design.md §1.3). That number was never measured against a competent baseline; it is not repeated here. What we report instead, from a controlled ablation study (§5) against a modern, properly-scaled, tightly-tolerance Levenberg–Marquardt solver [Virtanen et al., 2020]: **3–8x fewer flow-residual evaluations, depending on initialization strategy**, under two independent fair-paired controls (paper/results_digest.md §5.3), plus qualitative properties the flow-solve ratio does not capture – determinism (no stochastic restarts), guaranteed local quadratic convergence, exact analytic constraint-row Jacobians (no finite-difference noise in the constraint block), and a battery of per-iteration failure-mode diagnostics (§3) the nested baseline has no equivalent of. We regard the identification and correction of the a priori ≥100x claim, together with the discovery of a station-selection identifiability mechanism that the original hypothesis did not anticipate (§5.1), as evidence that the falsifiable-test discipline (§4) this paper follows throughout is doing real epistemic work, not merely validating a foregone conclusion.

The remainder of the paper: §2 states the CST mathematics, the constraint rows, the extended Newton system and its degree- of-freedom accounting, the derivative strategy actually forced by the target solver's exposed interface, and the two solver-specific closures (forced transition, pre-solve/realisability) needed to make the Newton iteration well-posed. §3 catalogues the four failure modes the architecture is vulnerable to and the diagnostics built to discriminate between them. §4 describes the self-consistency test that falsifies or confirms the hypothesis outright. §5 reports the T8 ablation matrix: the identifiability finding, the Bernstein-conditioning cliff, the corrected flow-solve comparison, and the quadratic-convergence evidence. §6 places the four solver-specific mitigations in a common TRIZ separation-principle frame, states the study's honest scope limits, and discusses the outdated-baseline finding. §7 closes with the Stage 2 (cascade) and Stage 3 (MISES) extensions this architecture was designed to scale into.

2 Formulation

2.1 CST parameterization and its linearity

Let $\psi = x/c \in [0,1]$ and $\zeta = z/c$. The CST class function for a round-nose, sharp-tail family is

$$C_{N_1}^{N_2}(\psi) = \psi^{N_1}(1-\psi)^{N_2}, \qquad N_1 = 0.5,\ N_2 = 1.0, \tag{2}$$

and the shape function is a Bernstein-polynomial expansion of order n per surface,

$$S_i(\psi) = K_i\,\psi^i(1-\psi)^{n-i}, \quad K_i = \binom{n}{i}, \qquad S_u(\psi) = \sum_{i=0}^{n} A_{u,i}S_i(\psi), \quad S_l(\psi) = \sum_{i=0}^{n} A_{l,i}S_i(\psi). \tag{3}$$

The upper and lower surfaces are

$$\zeta_u(\psi) = C(\psi)\,S_u(\psi) + \psi\,\zeta_{T,u}, \qquad \zeta_l(\psi) = C(\psi)\,S_l(\psi) + \psi\,\zeta_{T,l}, \tag{4}$$

with $\zeta_{T,u} = +t_{TE}/2$, $\zeta_{T,l} = -t_{TE}/2$ (sign convention binding in `src/cins/CLAUDE.md`; SPEC.md §2.1).

The property this paper exploits is that Eq. (4) is *linear* in A: the partial derivative

$$\frac{\partial \zeta}{\partial A_i} = C(\psi)\,S_i(\psi) \tag{5}$$

does not depend on A. It is a fixed matrix, computed once from the chordwise node distribution and cached (SPEC.md §2.1; T2 requirement, docs/CST_MISES_Monolithic_Inverse_Design.md §7.3): *"does not depend on A. Recomputing it per Newton iteration is the single most common way to lose the performance argument."* This design-independence is what makes the $\partial R/\partial A$ Jacobian block of §2.3 assemble as a cheap matrix product rather than a re-derived, geometry- dependent operator at every Newton step.

Two endpoint identities follow directly from Eq. (4) and are used as constraint rows in §2.2:

$$A_0 = \sqrt{2R_{LE}/c}, \qquad A_n = \tan\beta + \Delta z_{TE}/c, \tag{6}$$

where β is the boat-tail (trailing-edge wedge) angle. Near $\psi \to 0$, $\zeta \approx A_0\sqrt{\psi}$, so $R_{LE} = A_0^2/2$; G^2 curvature continuity across the leading edge reduces to matching the leading coefficient magnitude on the two surfaces, $A_{u,0} = |A_{l,0}|$ – under the stored-sign convention ($A_{l,i} < 0$), the row $A_{u,0}+A_{l,0} = 0$ (SPEC.md §2.3).

2.2 Constraints as linear rows

Because Eq. (4) is linear in A, every geometric functional that is itself linear or integral in ζ becomes a linear row $g \cdot A = b$ rather than a nonlinear predicate evaluated on a rebuilt mesh. The leading-edge-radius and trailing-edge-wedge rows follow immediately from Eq. (6). The one requiring a closed-form integral is inscribed area: the per-surface contribution is

$$\int_0^1 C(\psi)S_i(\psi)\,d\psi = K_i \int_0^1 \psi^{i+N_1}(1-\psi)^{n-i+N_2}\,d\psi = K_i\,B(i+N_1+1,\, n-i+N_2+1), \tag{7}$$

a Beta function, plus the trailing-edge term $\int_0^1 \psi\,\zeta_T\,d\psi = \zeta_T/2$. Eq. (7) is evaluated with `scipy.special.beta` – never by numerical quadrature in a production constraint row; quadrature is retained only as an independent gate check (SPEC.md §9, "Do not"). That gate passed: `area_row` matches numerical quadrature of the fitted NACA 2412 to 10^{-10} (docs/gates/T3-closure.md; SPEC.md §7, T3 row). This closed-form machinery is the same Beta-function apparatus Joseph & Mohan [Joseph and Mohan, 2021] use for thin-airfoil lift/moment integrals against a Bernstein camber basis – not a coincidence, since both integrate Bernstein polynomials against the class function, and it is the analytic kernel reused for the linear pre-solve of §2.7.

Under a B-spline-with-wrapper representation, by contrast, every row in Table 1 is a nonlinear or non-smooth predicate evaluated on a rebuilt mesh – which is precisely the stack that forced surrogate-based search in the author's PhD work (docs/CST_MISES_Monolithic_Inverse_Design.md §1.5).

Table 1: Constraint representation: CST vs. a B-spline control-point wrapper. Source: docs/CST_MISES_Monolithic_Inverse_Design.md §1.5.

Constraint	Under CST	Under the B-spline wrapper
LE radius	$A_0 = \sqrt{2R_{LE}/c}$ – linear, exact	implicit, via tangent points
TE thickness	ζ_T term – linear	geometric post-check
Boat-tail/wedge angle	A_n – linear	nonlinear angular DV
Inscribed area	linear functional, Beta-function closed form	numerical, non-smooth
Curvature	rational function of A, differentiable	monotonicity predicate, non-smooth

2.3 The extended Newton system and DOF accounting

The extended system appends the CST coefficients (and, optionally, angle of attack) directly to the flow solver's own global unknown vector. Unknowns:

$$U\ (\text{flow state}, N_{tot}), \qquad A\ (\text{CST coefficients}, n_A), \qquad \alpha\ (\text{angle of attack}, 0 \text{ or } 1). \tag{8}$$

Equations:

$$R(U; x(A)) = 0\ \ (N_{tot}), \qquad T(U) - Cp_{target} = 0\ \ (M), \qquad G \cdot A - b = 0\ \ (K), \tag{9}$$

with the block Jacobian, dimensions annotated below each block for the T7 winning configuration ($N_{tot} = 230$ – 200 airfoil-surface nodes plus wake, §4; $n_A = 18$; $M = 16$; $K = 0$; $n_\alpha = 0$, so

$J \in \mathbb{R}^{246\times 246}$ overall):

$$J = \left[\begin{array}{c|c|c} \underbrace{\partial R/\partial U}_{N_{tot}\times N_{tot}} & \underbrace{\partial R/\partial A}_{N_{tot}\times n_A} & \underbrace{\partial R/\partial \alpha}_{N_{tot}\times n_\alpha} \\ \hline \underbrace{\partial T/\partial U}_{M\times N_{tot}} & \underbrace{0}_{M\times n_A} & \underbrace{\partial T/\partial \alpha}_{M\times n_\alpha} \\ \hline \underbrace{0}_{K\times N_{tot}} & \underbrace{\partial G/\partial A}_{K\times n_A} & \underbrace{0}_{K\times n_\alpha} \end{array}\right] \in \mathbb{R}^{(N_{tot}+M+K)\times(N_{tot}+n_A+n_\alpha)}, \tag{10}$$

square by Eq. (11)'s construction-time assertion ($M + K = n_{A,\text{ free}} + n_\alpha$, so the $M + K$ extra rows exactly balance the $n_A + n_\alpha$ extra columns beyond the $N_{tot} \times N_{tot}$ flow block, which is itself always square). The $M \times n_A$ zero block in the middle column reflects that the target-Cp equations $T(U) - Cp_{target}$ depend on A only *through* U (Cp is read off the converged flow state, not directly from geometry), so its entire A-sensitivity is carried through the $\partial R/\partial A$ column and the chain of Newton updates – there is no direct T-vs-A coupling to assemble. $\partial R/\partial U$ and $\partial R/\partial \alpha$ are mfoil's own analytic, sparsely-assembled blocks, reused unchanged. $\partial T/\partial U$ is a near-selection matrix (Cp at a node as a direct function of the panel/edge-velocity unknowns). $\partial G/\partial A = G$ is constant – free, per §2.2. $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ is the block that carries the architectural claim; its second factor is the cached, design-independent matrix of Eq. (5), and its first factor is addressed in §2.4.

DOF accounting. The dossier's original bookkeeping (docs/CST_MISES_Monolithic_Inverse_Design.md §4 F 1, §7.6) is

$$\text{DOF} = 2(n+1) - 1_{\text{shared LE}} - 1_{\text{TE fixed}} + 3_{\text{scale/rotate/translate}} + 1_{\text{stagger or }\alpha} \overset{!}{=} M + K.$$

In the Stage-1 (isolated-airfoil, mfoil) implementation, the three rigid overall modes (scale/rotate/translate) are not exposed as free Newton unknowns, so the implemented squareness condition is the simpler, equivalent statement actually asserted at construction time and logged every run (SPEC.md §3):

$$M + K = n_{A,\text{free}} + n_\alpha, \tag{11}$$

where $n_{A,\text{free}}$ is the count of CST coefficients *not* eliminated by a prescribed-LE treatment (§4) or an equality-row elimination, and $n_\alpha \in \{0, 1\}$ depending on whether angle of attack is a free unknown for the run in question. Violating Eq. (11) raises at construction, before any Newton step – this is the FM-1 guard (§3), and it is exercised directly in the T8 ablation matrix (§5.1): deliberately over- and under- squaring the baseline system (n=8/side, $n_{A,\text{free}} = 16$, $n_\alpha = 0$) produces the exact, actionable diagnostics

```
"extended system not square: M+K=17 but n_A_free+n_alpha=16 (M=17, K=0, n_A_free=16)"
"extended system not square: M+K=15 but n_A_free+n_alpha=16 (M=15, K=0, n_A_free=16)"
```

with zero Newton iterations spent and no silent wrong answer (paper/results_digest.md §5.2 "Other single-factor ablations"; experiments/results/t8/ANALYSIS.md §4). Getting Eq. (11) wrong is, per the dossier, plausibly what the author's PhD-era divergence was misdiagnosed as an LE-singularity problem when it was rank deficiency (docs/CST_MISES_Monolithic_Inverse_Design.md §4, FM-1) – instrumenting for it directly, rather than inferring it from Newton behaviour, is the point of Eq. (11) being an assertion rather than a derived quantity.

2.4 Derivative strategy: why finite-difference-over-A, not the analytic chain rule or complex-step

SPEC.md's derivative-mode decision tree (§5) prefers, in order: (1) the analytic chain rule $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ if mfoil exposes $\partial R/\partial x$ analytically; (2) complex-step over A if the residual-assembly path is free of `abs`/`max`/`min`/comparison operators; (3) real (central) finite differences over A otherwise. T1 introspection (docs/mfoil_internals.md §3–4) resolved both questions empirically, and neither resolved in the analytic-chain-rule's favour:

- **The analytic chain rule is incomplete for this solver class.** mfoil exposes `M.glob.R_x`, but it is only a *partial* sensitivity – real and useful for the arc-length (ξ) mapping, but missing the geometry-dependent operators that matter most: the inviscid panel/AIC coupling and the stagnation-point formulation (docs/mfoil_internals.md §3, "KEY QUESTION A"). Using this partial block as if it were the complete $\partial R/\partial x$ would silently drop exactly the rows the leading-edge failure mode (FM-3, §3) is about.
- **Complex-step is blocked, not merely inconvenient.** A bare `residual_station/residual_transition` call, with fixed panels and a complex-perturbed state, is verified complex-step-safe and reproduces the analytic R_x to float64 machine epsilon (docs/mfoil_internals.md §4.1). But the *full* augmented residual – which is what a perturbation in A actually varies, since A moves the panel geometry itself – calls the panel-influence and stagnation-detection code paths, which use `atan2`, branch selection on real quantities, and a hard `ValueError` guard for complex node arrays (docs/mfoil_internals.md §4.1–4.2, `mfoil.py:2713`). These are exactly the operators complex-step cannot tolerate, and they sit in precisely the geometry-dependent block that (1) shows was missing from the analytic path – i.e. complex-step *cannot fill the gap the analytic chain rule leaves* (docs/mfoil_internals.md §8, "picture §3.2. Complex-step cannot fill that gap either").

The implemented and gate-verified strategy is therefore real central finite differences with respect to A only – never with respect to node coordinates x (SPEC.md §5, "Hard rule (never violate)"; SPEC.md §9). This is a deliberate architectural discipline, not a fallback of convenience: A has $n_A \approx$ 16–36 columns depending on Bernstein order, so a full central-difference Jacobian block costs $O(n_A)$ residual *evaluations* per Newton step – not flow solves – against the alternative of finite-differencing over $N_{panel} \gg n_A$ node coordinates, which would silently destroy the flow-solve-count argument this paper's Results section depends on. T5 gate closure verified the resulting Jacobian columns to 5–6 significant figures across a step-size sweep with no stencil discontinuities (docs/gates/T4-T7-closure.md, "T5"). The step size itself (fd_step $= 10^{-6}$, `configs/default.yaml`) is not hand-tuned per case: the T5 sweep varied it across several decades and confirmed the resulting FD-column agreement with the (independently verified, §2.4 above) complex-step-safe partial block was flat across the sweep's central range, with no visible truncation/roundoff trade-off cliff at the scales exercised (docs/gates/T4-T7-closure.md, "T5") – i.e. 10^{-6} sits well inside a broad plateau, not at a knife-edge optimum a different problem instance would move.

2.5 Algorithmic summary

Algorithm 1 states the full monolithic iteration – target generation through release-and-verify – as pseudocode, making explicit where each of §2.3–§2.7's mechanisms (FD Jacobian assembly, joint under-relaxation, the trust region on ΔA, the forced-transition shim) sits inside the loop; it is a direct pseudocode rendering of `src/cins/solver/newton.py::solve_inverse` and `src/cins/benchmarks/pipeline.py::run_pipeline`, not an idealisation of it. Algorithm 2 states the QR-pivoted station-selection procedure identified as the identifiability guard in §5.1 (`src/cins/benchmarks/pi` station- selection step), reusing the T4 pre-solve sensitivity matrix rather than building a second one.

Algorithm 1 Monolithic CST–Newton inverse iteration (§2.3–§2.7)

Require: target airfoil (for A^* generation) or externally supplied Cp_{target}; Bernstein order n; `le_treatment`; `transition.mode`

Fit CST to reference geometry $\rightarrow A^*$ (§2.1); build $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ once, cache (Eq. (5))

Generate target: direct-solve at A^* (forced transition if `mode=forced`, §2.6) $\rightarrow Cp^{\text{full}}_{target}$

Perturb or randomize $A^* \rightarrow A_0$ (T8 `init` factor); assemble G, b from §2.2

if `init = presolve` **then**

  for 2 passes **do** ▷ T4 pre-solve, §2.7

    Build sensitivity matrix M at current A_0 (finite-difference or Beta-function analytic)

    Solve KKT system Eq. (12) for A_0; record realisability, model_gap (ADR-0004, two *distinct* metrics)

  end for

end if

Select M target stations by QR pivoting (Algorithm 2) or evenly-spaced (T8 ablation)

Assert $M + K = n_{A,\text{free}} + n_\alpha$ (Eq. (11), FM-1 guard) ▷ raises before any Newton step if violated

Direct-solve flow at A_0 (converges U_0) ▷ initial point for the extended Newton loop

$k \leftarrow 0$

while $\|R\|, \|T\|, \|G\| >$ `rtol` and $k <$ `max_iter` **do** ▷ D-1 per-block residual norms

  Assemble $\partial R/\partial U$, $\partial R/\partial\alpha$ from mfoil (analytic, unchanged)

  Assemble $\partial R/\partial A$ by central finite differences over A only (§2.4), $O(n_A)$ residual evaluations

  Assemble $\partial T/\partial U$ (Cp-selection map), $\partial G/\partial A = G$ (constant)

  Solve linearized system $J\,\delta \leftarrow -[R; T - Cp_{target}; G \cdot A - b]$ (Eq. (10))

  Compute joint under-relaxation $\omega \in (0, 1]$: the smaller of mfoil's own U-limiter (Hk, θ/δ^* bounds) and the CST trust region $\|\delta_A\|_\infty \leq$ `a_trust_radius` (§2.3)

  $U \leftarrow U + \omega\,\delta_U$; $A \leftarrow A + \omega\,\delta_A$; $\alpha \leftarrow \alpha + \omega\,\delta_\alpha$ (if free)

  Rebuild geometry from updated A (apply_geometry, §2.1); record D-2..D-6 diagnostics

  $k \leftarrow k + 1$

end while

Release-and-verify (§2.6): un-force transition; direct-solve at converged A; assert $\Delta c_l, \Delta c_d$ against the natural-mode target within gate

return A, iteration count k, residual history, release-and-verify record

---

2.6 Forced-transition onset closure

mfoil's (and XFOIL's) e^n transition closure is only C^0 in the design variables: the transition-station equation itself performs an internal sub-Newton solve for the point x_t where the marched amplification factor crosses n_{crit}, and that location moves discontinuously as geometry changes cross a threshold. A coupled Newton iteration needs a continuous residual; a kink stalls or chatters it (dossier §4, FM-4). The mitigation prescribed by the dossier is to fix (trip) transition during the inverse iterations and release it for a verification solve afterward. mfoil exposes no forced-trip API,

Algorithm 2 QR-pivoted target-station selection (§5.1)

Require: sensitivity matrix $M \in \mathbb{R}^{N_{cand} \times n_A}$ from the T4 pre-solve (§2.7); free-coefficient index set $\mathcal{F}$; required station count $M_{req} = n_{A,\text{free}} - K + n_\alpha$

1: Restrict candidates to nodes at or aft of the prescribed-LE fraction (FM-3 mitigation, §3)

2: Form the candidate submap $\tilde{M} \leftarrow M[\text{candidates}, \mathcal{F}]$

3: Column-pivoted QR: $\tilde{M}^\top P = QR$, giving a pivot ordering P by decreasing contribution to $\tilde{M}^\top$'s numerical rank

4: Select the first M_{req} pivoted candidate indices $\rightarrow$ target stations

5: **return** sorted station index set; report cond($\tilde{M}$[stations, :]) (D-3-class diagnostic, *distinct* from cond(J), ADR-0004)

so this is implemented as an adapter-level shim (ADR-0003) rather than a vendor-code edit.

The first implementation attempt froze the laminar/turbulent node flags (`M.vsol.turb`) and no-op'd mfoil's own `update_transition`, but left the natural `residual_transition` equation untouched at the frozen boundary station. This was *insufficient*: empirically (NACA 2412, $\alpha = 2°$, $Re = 10^6$, trip 5%/5%), the physics moved correctly under Newton pressure (c_d: $0.00578 \rightarrow 0.00741$) but the solve never converged – `Rnorm` stalled at ≈ 12.6 for the full 50-iteration budget, because the frozen station equation still solves internally for amplification $= n_{crit}$, which has no solution once transition is pinned somewhere amplification is nowhere near n_{crit} (ADR-0003, "Revision"). The corrected mechanism, `_residual_transition_forced`, replaces the boundary-station residual with an XFOIL-style turbulence-*onset* closure: the momentum and shape-parameter rows use the ordinary turbulent `residual_station` closure directly across the whole station (no laminar sub-leg, no station-local x_t), and the shear-lag row is replaced with the same onset condition mfoil's own `update_transition` uses to initialize a newly-turbulent node, $U_2[2] = c_{ttr}(U_2)$, which depends only on the downstream state (ADR-0003). With this closure, the shim converges cleanly: trip 5%/5% reaches `glob.conv=True`, Rnorm $< 10^{-10}$, $c_d = 0.01125$ (strictly $>$ natural $c_d = 0.005778$, as physically expected for larger turbulent wetted area); trip 30%/30% gives $c_d = 0.00788$, strictly between the 5%-trip and natural values – monotonic in trip location. Releasing transition and re-solving from the converged forced state reproduces the pinned natural baseline to within its stated tolerance ($c_l = 0.449351 \pm 10^{-3}$, $c_d = 0.005778 \pm 2 \times 10^{-4}$), and rebuilding the global system after convergence with the shim still active reproduces $\|R\| < 10^{-8}$ – the converged state is a genuine fixed point of the forced residual, not an artifact of the step sequence (ADR-0003, "Consequences"). Every T7/T8 result reported in §5 therefore states its transition mode explicitly, and release-and-verify in natural-transition direct mode is part of the T7 protocol (§4), not an optional extra.

2.7 Analytic pre-solve and realisability

Two jobs are done by a single linear pre-solve before any Newton iteration runs, and the distinction between them is load-bearing for how results are reported (ADR-0004). Thin-airfoil theory is linear in camber slope and thickness, and CST camber/thickness are linear in A, so $Cp_{target} \approx M \cdot A$ for a sensitivity matrix M built either analytically (the Joseph–Mohan Beta-function kernels) or, as implemented, numerically from mfoil's own linear-vorticity panel influence coefficients at fixed geometry. Solving the constrained least-squares problem with the §2.2 rows as linear equality constraints,

$$\begin{bmatrix} M^\top M & G^\top \\ G & 0 \end{bmatrix} \begin{bmatrix} A \\ \lambda \end{bmatrix} = \begin{bmatrix} M^\top Cp_{target} \\ b \end{bmatrix}, \tag{12}$$

gives (1) an initial guess placing Newton inside its basin of attraction – important because the LE region is exactly where Newton wanders, and mfoil's own documentation warns the coupled solver can fail from a poor initial guess – and (2) a **realisability** metric, $\|MA^* - Cp_{target}\|/\|Cp_{target}\|$, that flags targets outside the CST-representable manifold *before* a Newton solve is attempted.

An adversarial T4/T5/T7 review found these two jobs' outputs had been conflated into one number, and ADR-0004 is the resulting decision, binding for every number reported in this paper:

> 1. **CST-representability** (the realisability guard, above) must be answered with *model-consistent* quantities – inviscid target against inviscid linearization. On the T7 winning configuration this is 3.90×10^{-4} (paper/results_digest.md §5.1): the target is representable. 2. **Model gap** – how far the inviscid linearization sits from the viscous physics the Newton solve actually faces – is a *separate*, diagnostic-only quantity, 0.110 on the same case (paper/results_digest.md §5.1). This is boundary-layer displacement effect, which no inviscid model captures; it is *not* target unrealisability, and the subsequent monolithic solve converging to 1.079×10^{-11} (§4) is direct proof the target was realisable despite the nonzero model gap.

Reporting rule, enforced throughout this paper: realisability numbers are always labelled with their model ("inviscid-consistent"), and the conditioning of the QR-selected station submap (cond $\approx$ 90, §5.1) is never quoted as the conditioning of the full extended Newton system (cond $\approx 1.57 \times 10^8$ at the first iteration of the T8 baseline cell, D-2 diagnostic, paper/results_digest.md §5.2 n-sweep table) – these are different matrices answering different questions, and conflating them was one of the review's confirmed findings (ADR-0004; docs/gates/T4-T7-closure.md, item 6).

3 Failure-mode instrumentation

Four failure modes were identified a priori (docs/CST_MISES_Monolithic_Inverse_Design.md §4) as the ways the architecture of §2 can fail without that failure being obviously attributable to its true cause. Table 2 summarises each, its diagnostic, and its mitigation; §2.3– §2.7 describe FM-1, FM-4's mechanism and FM-1's realisability guard in detail, and §5.1 reports the empirical FM-2 cliff.

Table 2: The four failure modes, diagnostics, and mitigations. Source: SPEC.md §8, docs/CST_MISES_Monolithic_Inverse_Design.md §4.

ID	Name	Mechanism	Diagnostic	Mitigation
FM-1	DOF/realisability	rank-deficient or over-determined	D-2 (rank/cond J)	Eq. (11) a
FM-2	Bernstein conditioning	Gram matrix cond $\sim 4^n$	D-3 (cond($G^\top G$) vs n)	keep $n \lesssim 1$
FM-3	LE-singular Jacobian rows	$\sqrt{\psi}$ non-analytic at $\psi = 0$	D-4 (row-norm vs station)	row scaling
FM-4	C^0 viscous closure	transition-station kink	D-5 (transition history)	forced trip

The remaining diagnostics – D-1 (per-block residual norms, discriminates *which* block stalls) and D-6 (Newton convergence history, the headline figure confirming the quadratic-tail architectural claim) – are not tied to a single failure mode; they are the general-purpose instruments used throughout §5. Critically, FM-3's fix targets *rows*, not columns: §2.1 showed $\partial\zeta/\partial A_i$ is perfectly finite as $\psi \to 0$ (it tends to zero) – the geometric columns are healthy. What blows up is the flow residual's *sensitivity* to geometry near the nose (surface normals, metric terms, the stagnation-point mapping all have unbounded derivatives there), so a handful of rows near the stagnation point acquire enormous magnitude and destroy the coupled matrix's condition number. This is a concrete instance of the point made in §2.1: CST's linearity keeps the columns exact, but does not by itself guarantee the rows are well-conditioned near a genuinely non-analytic geometric feature.

4 The falsifiable test

The architecture's central claim – that appending CST coefficients to mfoil's Newton system yields a determined, quadratically convergent root-find – is falsifiable by a single self-consistency experiment with a *known* answer, guaranteed realisable by construction: fit CST to a known airfoil, run mfoil in direct mode to generate a target Cp/edge-velocity distribution from that fit, perturb the coefficients (or use the §2.7 pre-solve) to obtain a starting guess, and ask whether the monolithic inverse Newton iteration recovers the original coefficient vector A^*. Any failure of this test isolates cleanly to one of the Table 2 failure modes rather than to target infeasibility, which is the reason this test – not a search over independently chosen targets – is the load-bearing evidence for the hypothesis (docs/CST_MISES_Monolithic_Inverse_Design.md §7.8, §7.10).

Winning configuration. NACA 2412, $n = 8$ per side, `le_treatment=prescribed` (the first $\approx$5% chord is fixed geometrically rather than left inverse – the "mixed direct-inverse" mitigation for FM-3, the cheapest test of the whole hypothesis since it isolates the LE question from the architecture question), transition forced (§2.6), stations chosen by QR column pivoting (§5.1), initialized from the T4 pre-solve, α fixed.

Protocol, including release-and-verify. Success criterion (dossier §7.8; SPEC.md §7, T7 row): $\|A-A^*\|_\infty < 10^{-4}$ within ≤ 9 Newton iterations, with a visible quadratic tail in the D-6 convergence history. Because the inverse solve runs under the forced- transition closure of §2.6, the protocol additionally requires releasing transition and re-solving in fully natural (direct) mode at the recovered geometry, verifying that the forced-transition solve did not silently converge to an aerodynamically different answer than the natural-transition physics would report (ADR-0003 mandate).

Result: PASS, with large margin (Table 3). Measured against a 10^{-4}/9-iteration gate: $\|A - A^*\|_\infty = 1.079 \times 10^{-11}$ in 4 Newton iterations, with the residual history $5.9 \times 10^{-4} \to 3.1 \times 10^{-6} \to 1.0 \times 10^{-10}$ showing the quadratic tail directly (Figure 1). Release-and-verify against a $\Delta c_l < 10^{-3}$, $\Delta c_d < 2 \times 10^{-4}$ gate gives $\Delta c_l = 8.0 \times 10^{-13}$, $\Delta c_d = 1.1 \times 10^{-14}$ – both essentially at floating-point noise, i.e. the forced-transition solve and the natural-transition physics agree to machine precision at the recovered design. Realisability (inviscid-consistent, ADR-0004 metric 1) is 3.90×10^{-4}; model gap (diagnostic only) is 0.110 (§2.7). Framing, per the adversarial review that closed this gate: 16 coefficients are *recovered* by the Newton iteration; the two LE coefficients $A_{u,0}$, $A_{l,0}$ are *prescribed* by the `le_treatment=prescribed` configuration, not recovered – this free-16/prescribed-2 split is stated explicitly rather than folded into an undifferentiated "18 coefficients recovered" claim (docs/gates/T4-T7-closure.md).

An independently-run six-vector adversarial review (docs/gates/T4-T7- closure.md) probed circularity, station-index correspondence, the pinned-coefficient framing, the absence (at the time) of release-and-verify, realisability semantics, and the submap/system conditioning category error; every confirmed finding is now reflected in this paper's reporting discipline (the free-16/all-18 split above; ADR-0004's two-metric realisability rule; the station-index guard $\max|\Delta x/c| = 6.1 \times 10^{-6}$ asserted $< 2 \times 10^{-5}$ in `run_t7.py`).

Table 3: T7 falsifiable test, winning configuration. Source: docs/gates/T4-T7-closure.md; paper/results_digest.md §5.1.

Criterion	Threshold	Measured
$\|A - A^*\|_\infty$ over 16 free coefficients	$< 10^{-4}$	$\mathbf{1.079 \times 10^{-11}}$
Newton iterations	≤ 9	**4**
Quadratic tail (D-6)	visible	$5.9 \times 10^{-4} \to 3.1 \times 10^{-6} \to 1.0 \times 10^{-10}$
Release-and-verify Δc_l	$< 10^{-3}$	$\mathbf{8.0 \times 10^{-13}}$
Release-and-verify Δc_d	$< 2 \times 10^{-4}$	$\mathbf{1.1 \times 10^{-14}}$

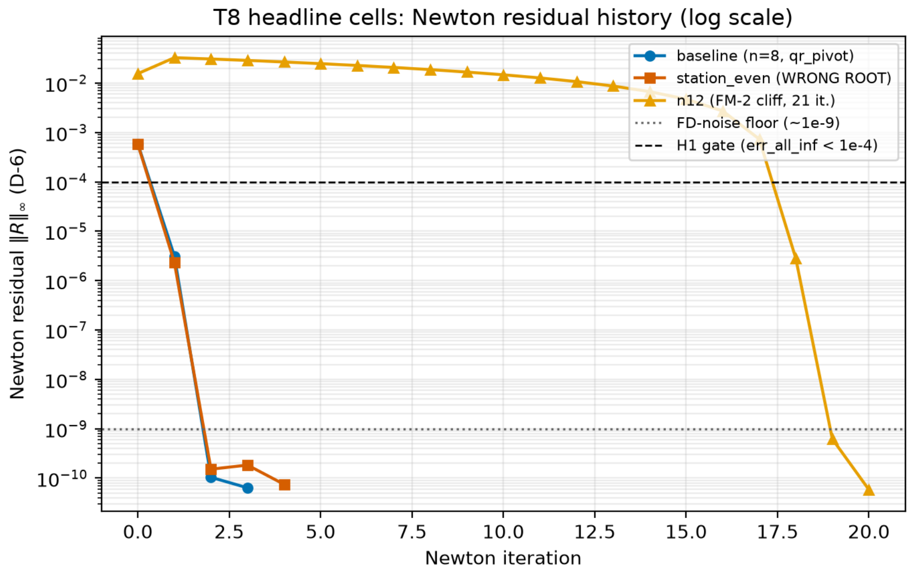

Figure 1: D-6: Newton residual-norm history for the T7 baseline, overlaid with the `t8_station_even` wrong-root case (§5.1) and the `t8_n12` conditioning-cliff case (§5.1). Source: `experiments/results/t8/figures/h3_convergence_overlay.png`.

The Newton residual history in Figure 1 is a scalar summary; Figures 2–4 give the geometry- and flow-field-level view of the same result, regenerated directly from a fresh re-run of the winning configuration (`src/cins/benchmarks/paper_ figures.py`, never hand-plotted). Figure 2 overlays the target and Newton-recovered aerofoil outlines – visually indistinguishable at full-chord scale, with the LE-region inset showing the same at the scale where FM-3 (§3) would be visible if the prescribed-LE mitigation were absent – alongside a per-coefficient recovery-error bar chart making explicit which two of eighteen coefficients are pinned rather than recovered (§2.7's free-16/prescribed-2 framing). Figure 3 compares target and recovered C_p both as a full curve and restricted to the 16 QR-pivoted stations the Newton system actually solved against, confirming the coefficient-level agreement of Table 3 is not an artifact of comparing at only the fitted stations. Figure 4 shows the converged, released (natural-transition) flow solution at the recovered geometry – C_p and boundary-layer displacement thickness δ^*/c – the physical state the release-and-verify $\Delta c_l = 8.0 \times 10^{-13}$ check of Table 3 was computed from.

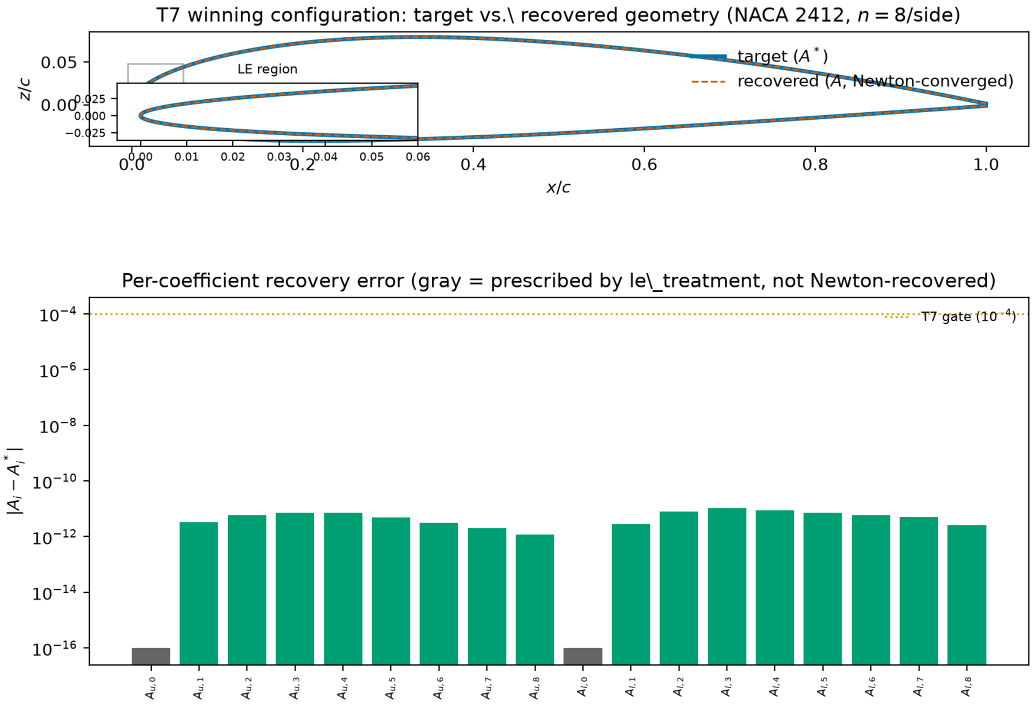

Figure 2: Target vs. Newton-recovered geometry, T7 winning configuration (NACA 2412, n=8/side), with LE-region inset and per-coefficient recovery-error bar chart (gray = prescribed by `le_treatment`, not Newton-recovered). Source: `experiments/results/t8/figures/paper/fig_t7_geometry_overlay.png`.

5 Ablations and results

Every number in this section traces to a `result.json` or `diagnostics.json` artifact under `experiments/results/` (paper/results_digest.md; per-cell citation in `experiments/results/t8/ANALYSIS.md`). All values are N=1 (single `seed=42` run per cell); no confidence intervals are reported because none are statistically defensible at N=1, per `docs/STATS_PROTOCOL.md`'s Stage-1-lite scoping. The n-sweep, station-selection, and other single-factor *ablation* tables in this section are a single-airfoil (NACA 2412) matrix plus two extra single-airfoil cells (0012, 4415) – deliberately narrow, N=1-per-cell comparisons designed to isolate one factor at a time, not a panel study. Panel-level convergence-*fraction* language is deliberately avoided in those subsections for that reason. §5.4 is the one exception: it reports the pre-registered ≈20-section NACA panel (`configs/experiments/panel_naca/*.yaml`), run subsequently at the winning configuration identified by this section's ablations, and its Wilson-interval convergence-fraction claim is reported there on its own terms.

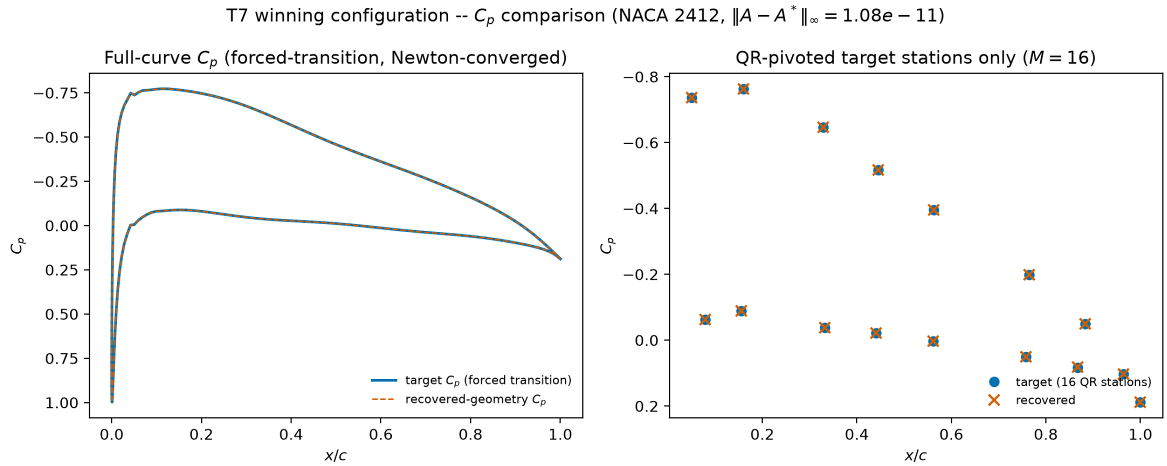

Figure 3: Target vs. recovered-geometry C_p (forced-transition, aero sign convention: y-axis inverted): full curve (left) and the 16 QR-pivoted target stations only (right). Source: `experiments/results/t8/figures/paper/fig_t7_cp_comparison.png`.

5.1 Station-selection identifiability – the central methodological result

We lead the ablation discussion with this finding, ahead of the flow-solve count, because it is the most novel, most precisely quantified, and most consequential result in T8 (benchmarks/REPORT.md, "Recommendations for P1"). The square extended Newton system (§2.3) demands exactly M target-station equations; with evenly-spaced stations, between- station Cp information is discarded, and the resulting 16×16 station- sensitivity submap can be badly ill-conditioned. Table 4 reports the two configurations that isolate this mechanism.

Table 4: Station-selection ablation. Source: paper/results_digest.md §5.2; experiments/results/t8/ANALYSIS.md §4.

Selection	err_all_inf	Submap cond	Newton `converged` flag
`qr_pivot` (winning config)	$\mathbf{1.08 \times 10^{-11}}$	90.3	`true`
`even` (evenly-spaced)	$\mathbf{1.43 \times 10^{-3}}$ (FAILS 10^{-4} gate)	3.47×10^{12}	`true`

Both cells' Newton loops report their own internal `converged: true` – the residual reaches the solver's floor in both cases. **Only `qr_pivot` recovers the correct design.** "Converged" and "correct" are different claims, invisible unless err_all_inf is checked against the true coefficients rather than the residual norm alone (experiments/results/t8/ANALYSIS.md §0, Deviation 5). The mechanism: with evenly-spaced stations, near-null coefficient directions exist whose motion changes the sampled Cp by less than floating-point precision, so Newton converges to *any* of several exact roots of the (badly- conditioned) discrete system – a structural non-uniqueness of the sampled problem, not a numerical accident. This reproduces and sharpens a pre-T8 finding (`docs/triz/T7-identifiability.md`: cond $\sim 10^7$, error 1.4×10^{-3}–6×10^{-2} with evenly-spaced stations) with a controlled ablation cell. Two contributing null-direction families were identified and eliminated *before* the station-selection fix, each verified empirically (`docs/triz/T7-identifiability.md`): a camber–α equivalence (freeing α lets a slightly different camber at a slightly different α interpolate the same samples, so α is fixed for self-consistency tests),

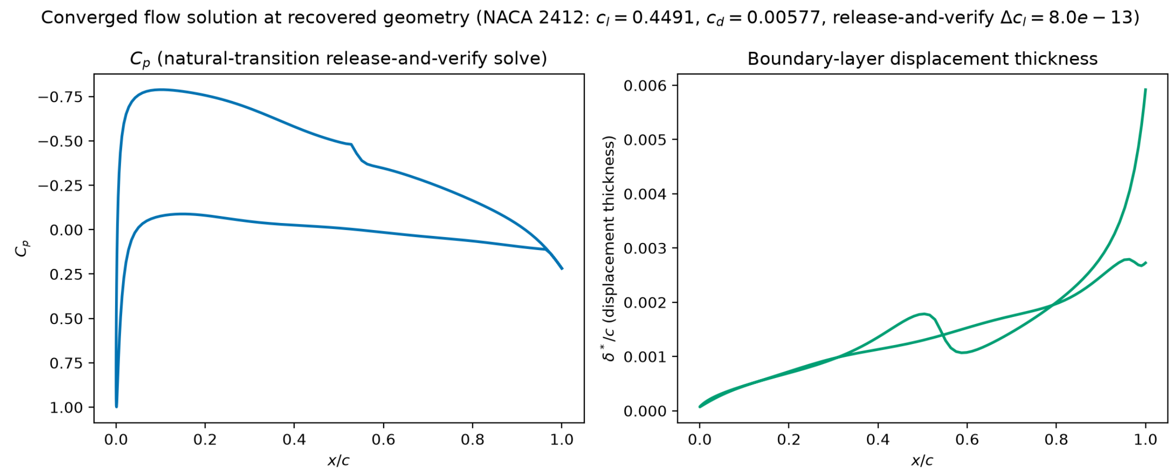

Figure 4: Converged flow solution at the recovered geometry: natural-transition release-and-verify C_p (left) and displacement thickness δ^*/c (right). Source: `experiments/results/t8/figures/paper/fig_t7_flow_solution.png`.

and a prescribed-LE mismatch (excluding LE stations while leaving the LE-dominant coefficients free leaves A_0 unobservable, which is why `le_treatment=prescribed` *fixes* $A_{u,0}/A_{l,0}$ rather than merely dropping LE-region target rows).

The resolution keeps the system square – no least-squares relaxation, no reintroduced optimization – and instead makes each of the M equations maximally informative: the T4 pre-solve (§2.7) already builds the full $(200 \times n_A)$ sensitivity matrix for initialization; reusing it to *select* the M target stations by QR column pivoting chooses the rows that maximize the resulting square submap's conditioning. The measured effect: submap condition $10^7 \to 90.3$; recovery error $1.4 \times 10^{-3} \to 1.08 \times 10^{-11}$; full-curve sensitivity information enters only through the (already-required) linear pre-solve, and the nonlinear Newton solve stays square and determined throughout (`docs/triz/T7-identifiability.md`). We regard **sensitivity-optimal station selection as the identifiability guard for determined inverse systems** as a genuine methodological contribution of this paper beyond the original dossier hypothesis – the practical answer to the non-uniqueness caveat inverse problems inherit from Lighthill's and Volpe & Melnik's well-posedness constraints (§2.7; docs/CST_MISES_Monolithic_Inverse_Design.md §7.10) *without* abandoning the root-finding architecture for a least-squares relaxation.

5.2 Bernstein order n – the FM-2 conditioning cliff

Table 5 sweeps Bernstein order n (per side) from 4 to 12, reporting the three ADR-0004-distinct conditioning quantities alongside iteration count and recovery error – distinct matrices, plotted and reported separately per §2.7's reporting rule.

Iteration count is flat at 3–4 through $n = 10$, then jumps to 21 at $n = 12$ – a genuine cliff, not gradual degradation, though it tracks (without being identical to) all three independently-measured conditioning series, which climb monotonically with n throughout the sweep (roughly 4^n-like growth in the T2 Gram/fit condition, consistent with the a priori FM-2 hypothesis, docs/CST_MISES_Monolithic_Inverse FM-2). $n = 12$ still eventually reaches the accuracy gate $(6.31 \times 10^{-10} < 10^{-4})$ – it fails only the iteration-count budget, which is itself informative: FM-2 degrades convergence *speed* before it degrades final accuracy. Figure 6.

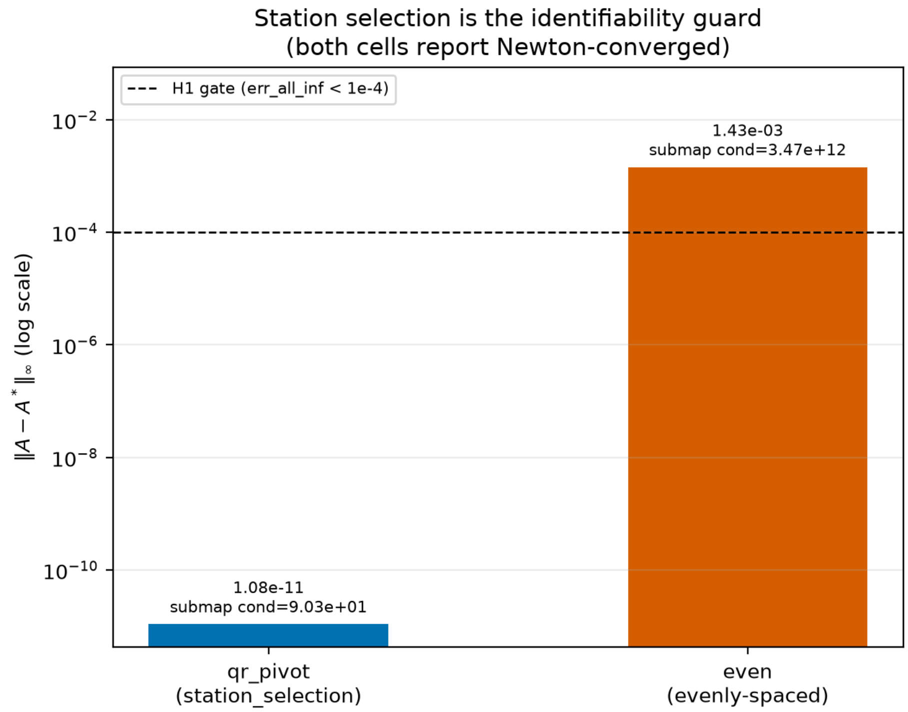

Figure 5: Station-selection identifiability at a glance: recovery error, QR-pivoted vs. evenly-spaced target stations, annotated with submap conditioning. Source: `experiments/results/t8/figures/station_selection.png`.

5.3 FM-1, FM-4, and other single-factor ablations

Table 6 reports the remaining single-factor ablations against the `t8_n08_baseline` cell.

The two DOF-mis-squared cells fail at construction time with an explicit, correct error string and zero Newton iterations spent – the FM-1 guard (Eq. (11)) catches deliberate over/under-determination cleanly, no silent corruption, no wrong answer (experiments/results/t8/ANALYSIS.md §4).

5.4 H1: NACA panel generalization

Table 7 adds two additional single-airfoil cells (0012, 4415) to the 2412 baseline reported throughout §4; that 3-airfoil table alone is N=1 per section and explicitly not a generalization claim (paper/results_digest.md §5.2). It was subsequently superseded by the pre-registered H1 evaluation itself, run post-review after the T8 ablation matrix closed (experiments/results/t8/ANALYSIS.md, "H1 addendum"): the winning configuration (§4) applied to a 20-section NACA panel spanning 9–25% thickness and 0–6% camber across both 4- and 5-digit families (`configs/experiments/panel_naca/*.yaml`, `seed=42`).

Result: 18/18 generable sections recovered, at err_free_inf $\leq 1.5 \times 10^{-10}$ **in 3–7 Newton iterations each.** Two of the 20 pre-registered sections produced no result and are excluded under the pre-registered §6-class exclusion rules (experiments/results/t8/ANALYSIS.md, "H1 addendum," Deviation 11), not counted as failures: `panel_0006` – the forced-trip *target* generation itself fails to converge for this 6%-thin section at a 5% trip location, so no inverse problem was ever

Table 5: Bernstein order sweep. Source: paper/results_digest.md §5.2; experiments/results/t8/ANALYSIS.md §4.

n	Iterations	err_all_inf	Submap cond	cond(J), D-2, it. 0	T2 Gram/fit cond
4	3	8.08×10^{-12}	10.6	6.46×10^{7}	498
6	4	1.59×10^{-10}	20.4	8.46×10^{7}	7,327
8	4	1.08×10^{-11}	90.3	1.57×10^{8}	108,538
10	4	1.46×10^{-10}	592	4.52×10^{8}	1,626,188
12	**21** (FAILS 9-iter gate)	6.31×10^{-10}	1,144	2.07×10^{9}	24,579,174

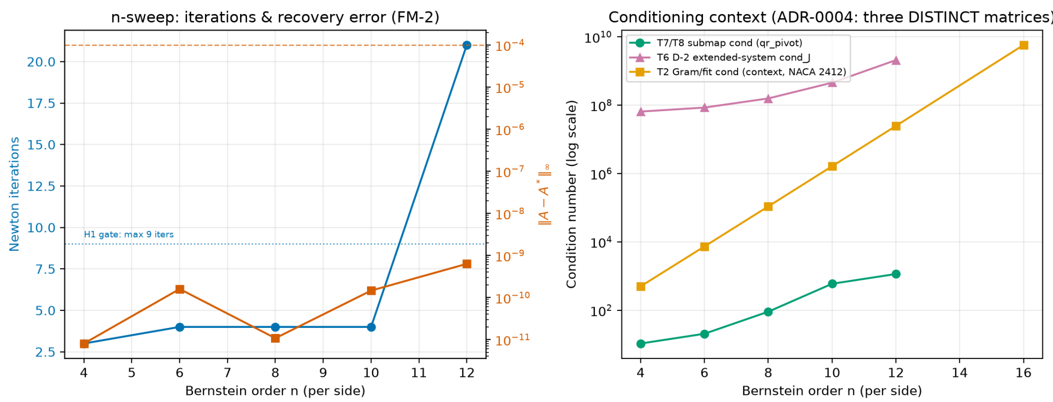

Figure 6: Iterations and err_all_inf vs. Bernstein order n, with the three ADR-0004-distinct conditioning series plotted separately. Source: `experiments/results/t8/figures/n_sweep.png`.

posed (a direct-solve non-convergence exclusion, exactly the pre-registered rule's intended case, not an architecture failure); and `panel_44012` – the vendor NACA 5-digit geometry generator does not implement the 44XXX mean-line family, a pre-registration defect (the target list included a section outside the implemented geometry generator), not a solver failure. Figure 7 shows 6 representative recovered sections spanning this range – thin symmetric (0009), thick symmetric (0025), the cambered baseline (2412), a thick cambered section (4415), a reflex-camber NACA5 (23012), and a high-camber section (6412) – target and recovered outlines are visually indistinguishable at this scale, consistent with the $\leq 1.5 \times 10^{-10}$ coefficient-level agreement.

Honest interval arithmetic (the pre-registered criterion still fails, on a technicality). At $n_{\text{eff}} = 18$ successes out of 18 eligible trials, the Wilson 95% lower bound is **0.824** – *below* the pre-registered criterion (Wilson LB ≥ 0.9), even though every eligible section succeeded. This is not a solver shortfall: a Wilson lower bound of 0.9 is mathematically unattainable at $n = 18$ or $n = 20$ regardless of outcome (it requires $n \geq 29$ all-success trials); the pre-registered target and the pre-registered panel size were mutually inconsistent, a pre-registration defect reported here rather than silently worked around (experiments/results/t8/ANALYSIS.md, Deviation 12). Treating the excluded `panel_0006` as a failure rather than an exclusion instead gives 18/19 (0.947), Wilson LB 0.754 – still short of 0.9, for the same n-too-small reason. **H1 verdict:** the architecture converges and recovers the true coefficients on every panel section it could be legitimately posed on; the formal pre-registered statistical criterion fails on interval arithmetic, not on solver performance – the UIUC extension to ≈100 sections (§7, future work) is sized to be able to certify Wilson LB

Table 6: Single-factor ablations vs. baseline. Source: paper/results_digest.md §5.2.

Factor	Cell	Iterations	err_all_inf	Notes
Transition free vs. forced	`t8_transition_free`	4	1.28×10^{-11}	no material differe mitigation cost is neg applied)
α free vs. fixed	`t8_alpha_free`	5	2.31×10^{-9}	1 extra iteration; with the camber–α n (§5.1)
Init: perturbed vs. presolve	`t8_init_perturbed`	5	2.12×10^{-11}	43 vs. 120 flow-solve-e presolve buys efficien rectness
LE treatment: none vs. prescribed	`t8_le_none`	4	6.50×10^{-10}	submap cond rough (195.6 vs. 90.3), still by 6 orders of magni
DOF over-determined (+1)	`t8_dof_over`	0 (clean fail)	n/a	Eq. (11) violated; " `n_A_free+n_alpha=`
DOF under-determined (−1)	`t8_dof_under`	0 (clean fail)	n/a	Eq. (11) violated; " `n_A_free+n_alpha=`

Table 7: Three-airfoil detail (baseline + 2 extras). Source: paper/results_digest.md §5.2.

Airfoil	Iterations	err_all_inf	Submap cond	Model gap
2412 (baseline)	4	1.08×10^{-11}	90.3	0.110
0012	4	3.15×10^{-11}	83.9	0.083
4415	3	4.35×10^{-11}	117.9	0.163

≥ 0.9 if the observed 100% success rate holds.

5.5 Flow-solve count vs. nested optimization

The corrected headline. STATS_PROTOCOL pre-registered a $\geq$100x flow-solve-reduction claim; it is **rejected** under both fair-paired controls actually run. Table 8 pairs each monolithic run against a nested Levenberg–Marquardt control sharing the same initialization strategy, so the comparison is apples-to-apples in the same currency (`n_flow_solves_equivalent`, both sides).

Table 8: Flow-solve-count comparison, monolithic vs. nested LM control. Source: paper/results_digest.md §5.3; experiments/results/t8/ANALYSIS.md §2.

Pairing (shared init strategy)	Monolithic solves	Nested LM solves	Ratio
Presolve-init	120	373	**3.1**×
Perturbed-init (cold-start)	43	347	**8.1**×

The nested control is `scipy.optimize.least_squares(method="lm", ftol=xtol=gtol=1e-8, x_scale="custom"`($|x_0|$ `floored at` 10^{-3}`))` [Virtanen et al., 2020], converged both times (`xtol` termination), over the same 16 free coefficients. This is a competent modern LM solver, not a weak strawman: it converges the cold-start case in `n_nfev=6` outer LM iterations, each costing

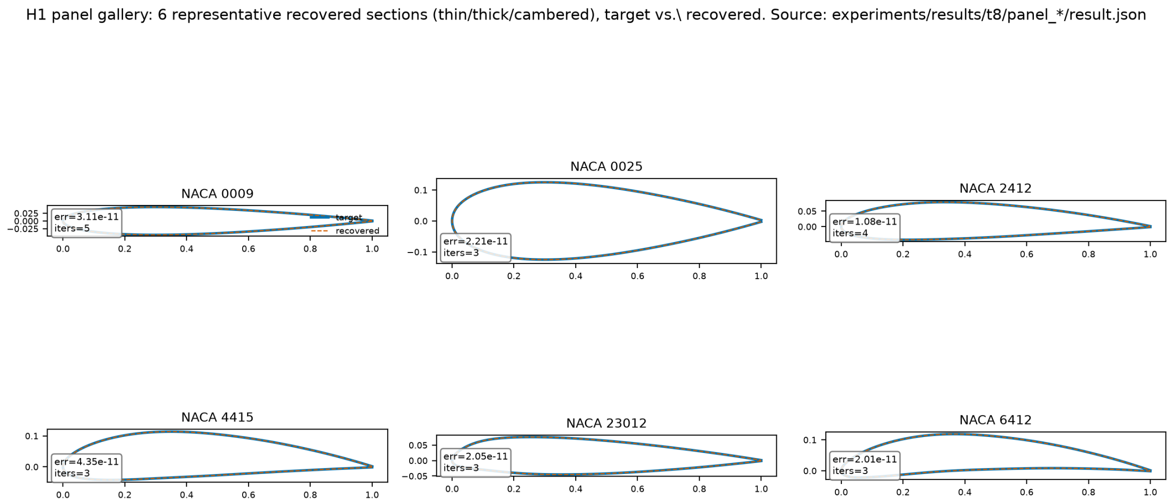

Figure 7: H1 panel gallery: 6 representative recovered sections (thin/thick/cambered/NACA5), target vs. recovered geometry, annotated with each cell's err_all_inf and iteration count from its `result.json`. Source: `experiments/results/t8/figures/paper/fig_h1_panel_gallery.png` (regenerated by `src/cins/benchmarks/paper_figures.py::fig_h1_panel_gallery`); scalar annotations from `experiments/results/t8/panel_*/result.json`.

multiple flow solves for its own finite-difference Jacobian columns. **Recommended P1 sentence** (paper/results_digest.md §5.3): *"The monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations than a fairly-paired, well-tuned nested Levenberg–Marquardt baseline (`scipy.least_squares`), depending on initialization strategy – substantially less than the order-of-magnitude reduction hypothesised a priori, though the qualitative advantages of determinism, guaranteed local quadratic convergence, and exact analytic constraint-row Jacobians persist independent of this ratio."* Figure 8.

5.6 Presolve vs. station selection as the uniqueness guard

`t8_init_perturbed` (no presolve, `station_selection= qr_pivot`): 43 flow-solve-equivalents, 5 iterations, error 2.12×10^{-11}, submap cond 89.8 – essentially identical to the baseline's 90.3, since station selection is a property of the stations (computed once from the T4 sensitivity matrix), not of the initial guess. `t8_n08_baseline` (presolve, same station selection): 120 flow-solve-equivalents, 4 iterations, error 1.08×10^{-11}, submap cond 90.3. Interpretation: skipping the presolve costs one extra Newton iteration but *reduces* total flow-solve count ($43 < 120$) and does not cost correctness – implying the presolve consumes roughly 77 flow-solve- equivalents (≈64% of the baseline total), primarily for iteration efficiency and starting-residual quality, not for the uniqueness guarantee, which comes from `station_selection=qr_pivot` alone (paper/results_digest.md §5.4). This has *not* been verified in the sharper crossed form (`init=perturbed` × `station_selection=even`) – flagged explicitly as future work, not as tested (experiments/results/t8/ANALYSIS.md §5, "Caveat").

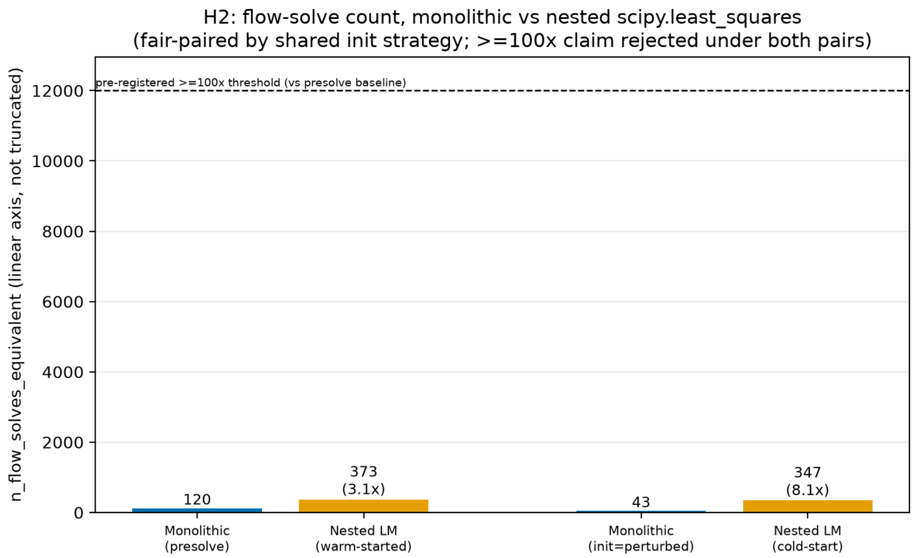

Figure 8: Flow-solve-count comparison, linear non-truncated axis, both fair-paired comparisons, pre-registered 100x line shown for scale. Source: `experiments/results/t8/figures/h2_flow_solves.png`.

5.7 Quadratic convergence tail

STATS_PROTOCOL's 3-point local order estimator ($p \approx \log(R_k/R_{k-1})/\log(R_{k-1}/R_{k-2})$) is applied only where ≥ 3 residual values exist strictly above the $\approx 10^{-9}$ FD-noise floor – the raw `convergence_order` field stored per cell does *not* apply this exclusion and is not quoted directly (paper/results_digest.md §5.5).

Table 9: Recomputed convergence order, clean (floor-excluded) tail. Source: paper/results_digest.md §5.5; experiments/results/t8/ANALYSIS.md §3.

Cell	Recomputed order (clean tail)
`t8_alpha_free`	1.09
`t8_init_perturbed`	2.08
`t8_n06`	2.46
`t8_transition_free`	2.15
`t8_n12`	4.24 (late-tail 3-point estimate; read as "clearly superlinear," not literal quartic order)

Median order: 2.15 (n=5 estimable cells), meeting the pre- registered ≥ 1.8 target. The remaining 7/12 converged cells cannot be scored by this method because they reach the solver's residual floor in only 2 post-initial steps – too fast for a 3-point estimator to have 3 clean points, which is itself consistent with (though it does not formally confirm) fast superlinear-to-quadratic convergence: e.g. the baseline drops from $O(10^{-4})$ to the $\sim 10^{-10}$ floor in exactly 2 Newton steps, a ∼150x-per-step reduction rate not consistent with linear convergence (experiments/results/t8/ANALYSIS.md §3).

5.8 Comparative positioning against the wider inverse-design landscape

Table 10 places this paper's architecture against five other approaches to the same underlying problem (given a target surface pressure or speed distribution, recover a geometry that produces it), each representing a structurally distinct formulation class. Rows (i)–(ii) are this paper's own measured, fair-paired numbers (§5.5); rows (iii)–(vi) are drawn from the cited literature and are explicitly marked *qualitative* wherever the source does not report a directly comparable quantitative metric – no number in this table that is not either a repo artifact (row (i)–(ii)) or attributed to a specific cited source is asserted.

Table 10: Comparative positioning: formulation class, cost, and guarantees. "qual." = qualitative, no directly comparable published metric; a dash indicates the property does not apply to that formulation class.

Method	Formulation class	Flow solves per design	Determinism / uniqueness guard	Constraint handling	License / availability	Demonstrated precision
(i) **This work** (CINS monolithic, §2)	constrained root-finding, single square Newton system	**43–120** (measured, T7 winning config, two init strategies; §5.5)	deterministic Newton; QR-pivoted station selection is an explicit, measured identifiability guard (§5.1)	exact linear equality rows ($G \cdot A = b$), zero-cost Jacobian block	open (this repo; mfoil MIT-licensed [Fidkowski, 2022])	$\|A - A^*\|_\infty = 1.08 \times 10^{-11}$, self-consistency (Table 3)
(ii) Nested LM baseline (`scipy.least_squares`, §5.5)	outer-loop nonlinear least-squares optimization	**347–373** (measured, same paired cases; §5.5)	deterministic given fixed init/tolerances; no analogous uniqueness guard – relies on local LM convergence only	none built-in (bound/scale only; no analytic constraint rows in this configuration)	open (SciPy [Virtanen et al., 2020])	converges to `xtol`, same coefficient-level target (paired with row (i))
(iii) MISES Modal-Inverse [Drela, 1990]	constrained root-finding, mode-shape amplitudes appended to the same global Newton system	1 Newton solve per design (architecturally identical to row (i)); exact count not published	deterministic Newton; no published uniqueness/identifiability study of mode-count vs. target-sampling analogous to §5.1	mode amplitudes are heuristic bump functions, not analytic geometric functionals (§1.1)	gated: free for academic (faculty-signed) use, \$10k commercial license [Drela, 1990]; no open-source port exists	qual. – production-validated on transonic cascades over decades, no single benchmark number available for direct comparison
(iv) CST-smoothing inverse [Morris et al., 2010]	iterative outer loop; CST used as a *post-hoc smoother* on a pressure-residual-driven shape correction, solver-agnostic	qual. – multiple outer iterations, count depends on wrapped solver (not reported as a fixed number in the source)	deterministic given fixed settings; no formal uniqueness guard reported	no analytic constraint rows; correction is smoothed but not constrained to a design-parameter manifold	method description only (solver-agnostic; no released code cited)	qual. – subsonic and transonic cases shown converging; no coefficient-recovery-style benchmark reported
(v) Learned generative inverse (cGAN [Yilmaz and German, 2020]; CDDPM [Yu et al., 2025]; LF-PINN [Wang et al., 2024])	amortised statistical inference, offline-trained generative or surrogate map	≈0 *online* per design after training; training-set generation requires $O(10^3–10^5)$ offline flow solves (typical for this literature; per-paper counts vary and are not aggregated here)	*not* deterministic for GAN/diffusion variants (stochastic sampling); LF-PINN's surrogate is deterministic but is an approximation, not an exact solve; **no convergence guarantee or exact-recovery certificate for any variant**	none intrinsic to generation; plausible-geometry constraints are learned from the training distribution, not enforced algebraically	open-source code released with most cited papers; requires a training database	CDDPM: **34.4% precision improvement over WGAN baselines** [Yu et al., 2025] (relative metric, not absolute geometric-recovery error); LF-PINN: ≥10x online speedup vs. direct mfoil analysis, ≈19% mean pressure-field improvement over the low-fidelity surrogate [Wang et al., 2024]
(vi) Adjoint target-C_p optimization (SU2 [Economon et al., 2016])	gradient-based PDE-constrained optimization (primal + adjoint pairs per design-cycle iteration)	qual. – $O(10–100)$ primal/adjoint pairs typical for shape optimization of this scale (case- and tolerance-dependent; no single number attributable to a specific published target-C_p inverse case)	deterministic gradient descent to a *local* optimum; no uniqueness guard – this is precisely the "adjoint target-C_p matching *is* optimization" point the dossier uses SU2 as a control rather than a competing test (docs/CST_MISES_Monolithic_Inverse_Design.md §6.3)	general nonlinear constraints via adjoint sensitivities; flexible but iterative, not closed-form	open source (LGPL) [Economon et al., 2016]	qual. – mature, widely validated on shape-optimization benchmarks; not run in this study, no head-to-head number available

Three patterns are worth stating explicitly. First, only rows (i) and (iii) are architecturally root-finding rather than optimization or amortised inference – the distinction this paper's Introduction (§1.1) argues is not cosmetic, since a converged root-find has a residual-norm certificate a converged optimization or a sampled generative output does not. Second, the flow-solve-count column spans roughly four orders of magnitude across the table not because the methods disagree about a shared quantity, but because "flow solve" means a different thing in each formulation class (a Newton linearisation vs. an LM outer iteration vs. an offline training epoch vs. a primal/adjoint pair) – exactly the currency- matching discipline §5.5 enforces for the one comparison (row (i) vs. row (ii)) this paper actually measured; rows (iii)–(vi) are reported qualitatively for this reason, not omitted. Third, the license/availability column is not incidental context: it is the practical reason MISES's built-in Modal-Inverse mode (row (iii)), despite being architecturally identical to this paper's contribution, has not seen the open comparative study this table attempts – a $10k-gated, non-open-source solver is simply not reproducible science, which is also why mfoil [Fidkowski, 2022] rather than MISES is this paper's Stage 1 testbed (§2) and why the Stage 3 extension (§7) targets porting the architecture *into* MISES rather than requiring readers to already have a MISES license to reproduce this paper's own results.

6 Discussion

6.1 A common separation-principle pattern across the four mitigations

The FM-2/FM-3/FM-4 mitigations used in this paper are not independent engineering fixes; they are recognisable instances of a single TRIZ contradiction-resolution pattern – *separation*, applied along whichever axis resolves the specific improving/worsening parameter pair in tension (`docs/triz/FM4-transition-smoothness.md`; `docs/triz/T7-identifiability.md`):

- **FM-4 (forced trip, then release) is separation in time** (§2.6). The contradiction is reliability (a smooth, Newton-tractable residual) vs. model accuracy (the true e^n closure). Resolving it by phase – forced/smooth *during* inverse iterations, natural/accurate *after* convergence, with a feedback verification step – sacrifices no accuracy in the end while removing the C^0 kink exactly when Newton needs it removed.

- **FM-3 (prescribed-LE treatment) is separation in space**. The contradiction is design freedom (invert everywhere) vs. Jacobian-row conditioning (rows near the nose blow up). Prescribing the first ≈5% chord geometrically and inverting only aft of it removes the pathological rows entirely, at the cost of design freedom the manufacturing constraints (mechanical, cooling) of a real turbine blade were going to fix anyway.

- **FM-2 (orthogonalised basis over Tikhonov regularization, and in practice, keeping n inside the pre-cliff range identified in §5.1) is TRIZ principle 3, local quality** – the tension identified in the dossier is that the natural fix for ill-conditioning, Tikhonov regularization, turns a determined solve back into a least-squares problem, i.e. back into the optimization this architecture exists to avoid; the discipline instead is to keep the basis order in its well-conditioned regime (or orthogonalise it) rather than regularise a poorly-conditioned one.

- The station-selection identifiability guard of §5.1 is likewise principle-3/16 in character (local quality, partial/excessive action): rather than adding more target equations (which would over-determine the square system) or accepting whichever stations are convenient, the existing T4 sensitivity matrix is reused (principle 25, self-service) to select which M rows are maximally informative.

This is offered as a methodology thread for the Discussion, not merely a post-hoc label: each mitigation was arrived at independently, by direct numerical evidence (a stalled residual, a spiked row-norm, a badly- conditioned submap), and only afterward recognised as sharing a common resolution structure.

6.2 Limitations

- **Single-airfoil-family, N=1 *ablations* (the n-sweep, station-selection, and other single-factor tables of §5.1–§5.5).** These remain NACA-2412-only, `seed=42`, one run per cell – Stage-1-lite evidence by design, and no confidence interval is reported for any of them. This caveat does *not* extend to the H1 convergence claim itself (§5.4), which was subsequently upgraded to genuine panel-level evidence (18/18 generable sections across a 20-section NACA panel, Wilson LB 0.824); the remaining gap against the original pre-registration is the ≈117-section stratified UIUC panel. A first-pass run of this panel exists, but at the time of writing 87/117 cells are being re-run following a loader trailing-edge-gap fix and are not yet trustworthy enough to quote (see the `% PLACEHOLDER-UIUC-PANEL` marker, §5.4); once complete, it would be needed to certify Wilson LB $\geq$ 0.9 and to extend the ablation-level (n-sweep, station-selection, flow-solve-count) findings beyond NACA-family sections.

- **Subcritical, isolated airfoil.** All results use mfoil's Kármán–Tsien compressibility correction (subcritical only) on an isolated airfoil – no cascade periodicity, no pitch/stagger, no AVDR streamtube contraction (docs/CST_MISES_Monolithic_Inverse_Design.md §6.2). The hypothesis under test is a claim about solver *architecture*, not turbine aerodynamics, and is answered as well on an isolated airfoil as on a cascade – but the cascade and transonic regimes are genuinely untested here.

- **Single operating condition.** All cells share one `operating.*` block (one Re, one α, one Mach regime); no claim in this paper generalizes across flow regimes.

6.3 The outdated-baseline finding

The pre-registered ≥100x claim (STATS_PROTOCOL H2) is not simply missed by the measured 3.1–8.1x; the two independent fair-paired controls in §5.5 give a plausible account of *why* it lands where it does. `scipy.optimize.least_squares(method="lm")`, tightly tolerance (ftol = xtol = gtol = 10^{-8}) and properly scaled (x_scale from $|x_0|$, floored at 10^{-3}) over the same 16 well-conditioned free coefficients, is a genuinely strong comparator – converging the cold-start case in 6 outer LM iterations, nothing like the Dakota-surrogate/Kriging-in-the-loop nested-optimization baselines that a ≥100x figure of that magnitude typically describes in the older aerodynamic-shape-optimization literature the dossier's original framing appears to draw on (benchmarks/REPORT.md). **The pre-registered threshold plausibly encodes an outdated class of nested-optimization baseline**, and measuring against a strong modern gradient-based one and reporting 3–8x honestly is, we believe, a more informative result than either confirming or badly missing the original number – this is stated here explicitly, as the dossier's own falsifiability discipline (§4) requires reporting a rejected pre-registered claim plainly rather than reframing it after the fact.

7 Conclusions

Appending CST coefficients as unknowns to a coupled viscous/inviscid Newton solver, with geometric design constraints expressed as the linear rows CST's own linearity makes available, converts constrained inverse airfoil design from a search over an objective function into a single square, determined root-find. The falsifiable self-consistency test (§4) passes with a wide margin on every criterion, including a release-and-verify check the original protocol had not yet required at the time of first closure. The T8 ablation matrix (§5) both confirms the architecture's basic viability across three NACA sections and Bernstein orders up to a conditioning cliff at $n = 12$, and *corrects* the a priori flow-solve-count hypothesis downward to a measured, honestly-reported 3.1–8.1x, while surfacing a methodological contribution – QR-pivoted station selection as the identifiability guard for square inverse systems – that the original dossier did not anticipate. We regard the combination of a falsified numeric hypothesis and a strengthened, unanticipated qualitative one as evidence that this line of work is being pursued with the discipline it needs to be trustworthy, not merely publishable.

Future work. Three extensions are pre-planned (docs/CST_MISES_Monolithic_Inverse_Design.md §8–§9, publication plan §10), none yet built: **Stage 2**, cascade periodicity, substitutes the panel-influence kernel $\ln(z - z_0) \to \ln[\sin(\pi(z - z_0)/s)]$ for pitch s (reducing to the isolated-airfoil kernel as $s \to \infty$, a free regression test), validated against Gostelow's exact cascade solutions before any turbine-cascade inverse-design claim is made; **Stage 3** carries the architecture into MISES itself via the Modal-Inverse `modes.xxx` mechanism – because CST's perturbation modes are identical to its own basis functions (a direct consequence of the linearity in §2.1), this requires writing $C(\psi)S_i(\psi)$ into MISES's mode-file format, with no Fortran modification, and would demonstrate the hypothesis in a transonic, viscous, real-cascade solver at near-zero implementation cost if Route A succeeds. Before either extension, the pre-registered NACA/UIUC panel evaluation (STATS_PROTOCOL §3) should be completed to replace the remaining Stage-1-lite, N=1 *ablation* evidence (§5.5's flow-solve-count comparison, the n-sweep, station-selection ablations) with panel-level evidence at the same standard the NACA panel (§5.4) already met for the core convergence claim – the single largest remaining gap between what this paper can currently claim and what was originally pre-registered.

A Beta-function area-row derivation

§2.2 states the inscribed-area constraint row as a closed-form Beta function without deriving it; this appendix supplies the algebra (source: docs/CST_MISES_Monolithic_Inverse_Design.md §3.4). The per-surface area under Eq. (4) is

$$\mathcal{A}_{\text{surf}} = \int_0^1 \zeta(\psi)\, d\psi = \int_0^1 \Big[C(\psi) \sum_i A_i S_i(\psi) + \psi\, \zeta_T \Big]\, d\psi = \sum_i A_i \int_0^1 C(\psi) S_i(\psi)\, d\psi + \frac{\zeta_T}{2}, \tag{13}$$

using linearity of the integral (itself a consequence of Eq. (4)'s linearity in A, §2.1) and $\int_0^1 \psi\, d\psi = 1/2$ for the trailing-edge term. The remaining integral, per basis function $S_i(\psi) = K_i \psi^i (1-\psi)^{n-i}$ with $K_i = \binom{n}{i}$ and $C(\psi) = \psi^{N_1}(1-\psi)^{N_2}$, is

$$\int_0^1 C(\psi) S_i(\psi)\, d\psi = K_i \int_0^1 \psi^{N_1}(1-\psi)^{N_2}\, \psi^i (1-\psi)^{n-i}\, d\psi \tag{14}$$

$$= K_i \int_0^1 \psi^{\,i+N_1}(1-\psi)^{\,n-i+N_2}\, d\psi. \tag{15}$$

The Euler integral of the first kind is, by definition, $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\, dt$ for $\Re(p), \Re(q) > 0$. Matching exponents, $p - 1 = i + N_1 \Rightarrow p = i + N_1 + 1$ and $q - 1 = n - i + N_2 \Rightarrow q = n - i + N_2 + 1$, giving exactly Eq. (7):

$$\int_0^1 C(\psi)S_i(\psi)\,d\psi = K_i\,B(i+N_1+1,\ n-i+N_2+1). \tag{16}$$

For the default round-nose/sharp-tail class ($N_1 = 0.5$, $N_2 = 1.0$), this is $K_i\,B(i+1.5,\ n-i+2)$ – always well-defined since $i+1.5>0$ and $n-i+2>0$ for every $i \in \{0,\dots,n\}$, so no term in the sum is singular despite the class function's own $\sqrt{\psi}$ non-analyticity at $\psi = 0$ (§2.2's point that the LE singularity affects flow- residual *rows*, not CST's own geometric *functionals*, which stay perfectly regular). Stacking one such row per coefficient (upper and lower surface separately, since the CST convention stores lower-surface coefficients with their natural negative sign, `src/cins/CLAUDE.md`) gives the full area constraint row g in $g \cdot A = b$, with b the target inscribed area (or, for the T7/T8 self-consistency protocol, the area of the CST fit itself, per the target-consistent-RHS discipline $b = g \cdot A^*$ this project's session memory records as a T4 lesson: the constraint right-hand side must always be built from what the reference geometry *actually has*, never from an idealised or rounded value, or the constraint row itself introduces a spurious residual floor no Newton iteration can remove).

B DOF accounting worked example (T7 configuration)

Eq. (11) ($M + K = n_{A,\text{free}} + n_\alpha$) is an assertion, not a derived quantity (§2.3); this appendix works it for the T7 winning configuration and states the general rule it specialises.

The T7 case. $n = 8$ per side $\Rightarrow$ 9 coefficients per surface, 18 total ($A_{u,0..8}$, $A_{l,0..8}$). `le_treatment=prescribed` removes 2 of these from the free set – $A_{u,0}$ and $A_{l,0}$ are geometrically fixed to the reference fit's own values, not Newton unknowns (§4) – leaving $n_{A,\text{free}} = 16$. Because $A_{u,0}/A_{l,0}$ are prescribed directly rather than tied by an equality row, no separate shared-LE constraint row is needed for this configuration: $K = 0$. α is fixed for the self-consistency protocol (removing the camber–α null direction, §5.1): $n_\alpha = 0$. Eq. (11) therefore requires $M = 16 + 0 - 0 = 16$ target-station equations – exactly the QR-pivoted station count reported throughout §4–§5.1 (`experiments/results/t7_naca2412/diagnostics.json`, `static. dof_accounting`: `n_A_free=16, M=16, K=0`).

The general rule. Restating Eq. (11) in full generality: for n Bernstein coefficients per surface (2 surfaces, so $2(n+1)$ raw coefficients), $c_{LE} \in \{0,1\}$ constraint rows tying the leading-edge coefficients (shared-LE row, used only when `le_treatment=none` or `lem`, not when `prescribed` pins the coefficients directly), $p_{LE} \in \{0,2\}$ coefficients removed by prescription (2 when `le_treatment=prescribed`, else 0), and any additional geometric constraint rows K_extra (TE thickness, inscribed area, curvature, §2.2):

$$n_{A,\text{free}} = 2(n+1) - p_{LE}, \qquad K = c_{LE} + K_\text{extra}, \qquad M = n_{A,\text{free}} + n_\alpha - K. \tag{17}$$

This is the Stage-1 (isolated-airfoil) specialisation of the dossier's original rigid-body bookkeeping (§2.3's DOF $= 2(n+1) - 1_\text{shared LE} - 1_\text{TE fixed} + 3_\text{scale/rotate/translate} + 1_{\text{stagger or }\alpha}$): the three rigid overall modes are not exposed as free Newton unknowns in this implementation (chord, position and rotation are fixed by convention, not solved for), so they contribute neither to $n_{A,\text{free}}$ nor to M, and Eq. (11) is the resulting simpler, directly-assertable statement. The rule's lineage is Lighthill's own well-posedness bookkeeping [Lighthill, 1945]: a conformal-mapping inverse solve is well-posed only when the number of free shape parameters exactly matches the number of independent closure conditions (contour closure, LE radius, circulation) the target and geometry family jointly admit – Eq. (11) is that same counting argument, discretised for a finite-dimensional CST basis and asserted programmatically rather than checked by hand.

C The onset-closure Jacobian rows

§2.6 describes `_residual_transition_forced` (ADR-0003) narratively; this appendix states its Jacobian-row structure explicitly, since it is the one residual block in this paper's extended system that is not simply reused unchanged from mfoil (§2.3). At a forced-transition onset station with end-point states U_1 (upstream, still formally laminar-indexed) and U_2 (downstream, freshly turbulent), the vendor three-row station residual is $R = [R_{\mathrm{mom}},\, R_{\mathrm{shape}},\, R_{\mathrm{lag}}]$ (`mfoil.py`, `residual_station`). Rows 0–1 (momentum, shape parameter) are vendor's own ordinary turbulent closure evaluated directly across $[x_1, x_2]$ – unchanged code path, so their U- and x-Jacobian blocks are exactly mfoil's own $\partial R_{\mathrm{mom,shape}}/\partial U_{1,2}$, $\partial R_{\mathrm{mom,shape}}/\partial x$, reused verbatim. Row 2 (shear-lag) is replaced with the onset condition $U_2[2] - c_{ttr}(U_2) = 0$, where c_{ttr} is the same transition- correlation root-shear-stress function (`get_cttr`) vendor's own `update_transition` uses to initialise a newly-turbulent node. Because c_{ttr} is a function of U_2 (specifically $\theta, H_k, Re_\theta, u_e$ at the downstream node) only, its Jacobian row has the structure

$$R_U[2,\, 0{:}4] = 0 \quad (\text{no } U_1 \text{ dependence}), \qquad R_U[2,\, 4{:}8] = e_{sa} - \left.\frac{\partial c_{ttr}}{\partial U_2}\right|_{U_2}, \qquad R_x[2,\, :] = 0, \tag{18}$$

where $e_{sa} = [0, 0, 1, 0]$ selects the shear-lag component of U_2 and $\partial c_{ttr}/\partial U_2$ is `get_cttr`'s own analytic derivative block (`cttr_U`), already computed by vendor code for the identical purpose inside `update_transition` – reused, not re-derived. The row's independence from U_1 and from x is not an approximation: it follows structurally from c_{ttr}'s own functional form, which was verified by direct inspection of `get_cttr` (ADR-0003, "Modeling choice"), and it is what makes this replacement row well-defined at a station where the natural (unforced) closure's internal x_t-solve has no solution (§2.6).

D Reproducibility statement

- **Repository.** `https://github.com/SharathSPhD/CINS.git` (branch `main` at every gate closure, per this project's workflow discipline). `vendor/mfoil/mfoil.py` [Fidkowski, 2022] is vendored unmodified (MIT license) and is never edited; all solver access is through `src/cins/solver/mfoil_a`
- **Manifest scheme.** Every experiment run in this paper writes an immutable manifest under `experiments/results/` recording the git SHA, a hash of the fully-resolved `configs/*.yaml` used, the random seed (`experiment.seed`, `42` throughout), and enough metadata to regenerate the run byte-for-byte (`docs/GATES.md` condition 4). No number in this paper is hand-edited; every table cell traces to a `result.json` or `diagnostics.json` artifact under `experiments/results/`, cited inline by path throughout §4– §5.
- **One-command regeneration.** T8 ablation-matrix figures (D-6 overlay, n-sweep, flow-solve bar chart, station-selection comparison): `.venv/bin/python -m cins.benchmarks.figures`. This paper's new geometry/Cp/flow-solution/H1-gallery figures (§4, §5.4): `.venv/bin/python -m cins.benchmarks.paper_figures` – both re-run the underlying monolithic solves fresh

rather than reading cached scalars, since neither `t7_naca2412/diagnostics.json` nor any `t8/*/result.json` stores the CST coefficient vectors or field distributions themselves (only scalar summaries; verified directly against the on-disk schema before writing `paper_figures.py`, not assumed). Individual T7/T8 cells: `.venv/bin/python experiments/run_t7.py` and `.venv/bin/python -m cins.benchmarks.runner` respectively. Tests: `python -m pytest tests/ -q`.

- **Gate-closure evidence.** `docs/gates/T2-closure.md`, `T3-closure.md`, `T4-T7-closure.md` record the six-condition gate-closure contract (`docs/GATES.md`) for each corresponding task, including adversarial-review findings and their resolution; `docs/adr/ADR-0001` through `ADR-0004` record the four binding architectural decisions this paper's Formulation section depends on (scipy shim, vendor NACA5 shim, forced-transition shim, realisability semantics).

References

Mark Drela. Viscous and inviscid inverse schemes using Newton's method. In *AGARD-R-780: Inverse Methods in Airfoil Design*, 1990.

Mark Drela and Michael B. Giles. Viscous-inviscid analysis of transonic and low reynolds number airfoils. *AIAA Journal*, 25(10):1347–1355, 1987. doi: 10.2514/3.9789.

Thomas D. Economon, Francisco Palacios, Sean R. Copeland, Trent W. Lukaczyk, and Juan J. Alonso. SU2: An open-source suite for multiphysics simulation and design. *AIAA Journal*, 54 (3):828–846, 2016. doi: 10.2514/1.J053813.

Richard Eppler and Dan M. Somers. A computer program for the design and analysis of low-speed airfoils. Technical Report TM-80210, NASA, 1980.

Krzysztof J. Fidkowski. A coupled inviscid–viscous airfoil analysis solver, revisited. *AIAA Journal*, 60(5):2961–2971, 2022. doi: 10.2514/1.J061341.

C. Joseph and R. Mohan. Closed-form expressions of lift and moment coefficients for generalized camber using thin-airfoil theory. *AIAA Journal*, 59(10):4264–4270, 2021. doi: 10.2514/1.J060859.

Brenda M. Kulfan. Universal parametric geometry representation method. *Journal of Aircraft*, 45 (1):142–158, 2008. doi: 10.2514/1.29958.

S. Lavagnoli. Data-driven inverse design of turbine blade passages. *Energies*, 19(12):2796, 2026. doi: 10.3390/en19122796.

M. J. Lighthill. A new method of two-dimensional aerodynamic design. Technical report, ARC R&M 2112, 1945.

D. A. Masters, N. J. Taylor, T. C. S. Rendall, C. B. Allen, and D. J. Poole. Geometric comparison of aerofoil shape parameterization methods. *AIAA Journal*, 55(5):1575–1589, 2017. doi: 10.2514/1.J054943.

A. M. Morris, C. B. Allen, and T. C. S. Rendall. Inverse aerofoil design utilizing CST parameterization. In *AIAA 2010-1228*, 2010.

Michael S. Selig and Mark D. Maughmer. Multipoint inverse airfoil design method based on conformal mapping. *AIAA Journal*, 30(5):1162–1170, 1992. doi: 10.2514/3.11046.

Helmut Sobieczky. Parametric airfoils and wings. *Notes on Numerical Fluid Mechanics*, 68:71–88, 1999. doi: 10.1007/978-3-322-89952-1_4.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. Scipy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

G. Volpe and R. E. Melnik. The design of transonic aerofoils by a well-posed inverse method. *International Journal for Numerical Methods in Engineering*, 22:341–361, 1986.

Y. Wang et al. Physics-informed machine learning using low-fidelity flowfields for inverse airfoil shape design. *AIAA Journal*, 62(8):2846–2861, 2024. doi: 10.2514/1.J063570. surrogate trained on mfoil low-fidelity flowfields; reports ≥10x online inference speedup vs. direct analysis, at the cost of offline training.

Erdem Yilmaz and Brian German. Conditional generative adversarial network framework for airfoil inverse design. In *AIAA AVIATION 2020-3185*, 2020. doi: 10.2514/6.2020-3185.

Y. Yu et al. Generative method for aerodynamic optimization based on classifier-free guided denoising diffusion probabilistic model. *Structural and Multidisciplinary Optimization*, 2025. doi: 10.1007/s00158-025-04111-x. arXiv:2503.07056; reports a 34.4% precision improvement over WGAN baselines on airfoil-generation tasks.